

Keshav Rajput

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EDUCATION

New York University, Tandon School of Engineering

MS, Computer Science, May 2026. GPA 3.67/4.0

IIIT Delhi

B.Tech, Computer Science and Applied Mathematics; minor in Economics, 2020 - 2024. CGPA 8.19/10

Coursework: Design and Analysis of Algorithms II, Principles of Database Systems, Open Source and Professional Software Development, Distributed Systems, Operating Systems, Computer Networks

PROJECTS

Atlas (in progress) | C++23, SDL_GPU, HLSL, WebAssembly, ENet github.com/happyc0der/atlas-engine

- Solo C++23 engine for map-based strategy games, 25 gated milestones: SDL_GPU renderer, hot-reload assets, versioned serialization, 21 ADRs.
- Deterministic fixed-tick simulation with state hashing, replay and lockstep over ENet, proved hash for hash across two processes; million-cell tick in 694 us.
- Mods run as WebAssembly with no clock, filesystem or RNG; 1045 tests, CI on macOS, Linux and Windows with ASan/TSan and gated benchmarks.

mldsa-auth | C11, liboqs, libsodium, SQLite, CMake github.com/happyc0der/mldsa-auth

- Post-quantum mutual auth and session protocol in C11: ML-DSA-65 signatures over hybrid X25519 + ML-KEM-768, ChaCha20-Poly1305 records.
- 0.45 ms median handshake, 736 MiB/s record sealing; ASan/UBSan CI on Linux and macOS, 12 fuzz targets, 192 mutations nightly.
- Built mldsa-authd on it: login codes, device enrolment, revocation, key rotation, recovery codes; ProVerif model of the handshake.

doom-ports | C, eZ80 assembly, Monkey C, Connect IQ SDK, CEdev github.com/happyc0der/doom-ports

- From-scratch first-person raycaster for the TI-84 Plus CE in C and eZ80 assembly: 24 KB, 5 to 7 fps, all art generated at startup.
- Measured 122 eZ80 cycles per VRAM byte, fit a frame-time model (ceiling 6.2 fps) and moved the column fill to hand-written assembly.
- Garmin Venu X1 port in Monkey C, rebuilt around a 0.2 ms per-draw cost: 1 to 2 fps to 15.6 fps; submitted to the Connect IQ store.

ForgeLog | Kotlin, Jetpack Compose, Room, Hilt, DataStore github.com/happyc0der/ForgeLog

- Offline Android gym tracker in Kotlin (Jetpack Compose, Room, Hilt, MVVM); five permissions declared, none of them INTERNET.
- Rest timer on the lock screen survives process death; analytics for volume, estimated 1RM and personal bests; JSON backup and CSV export.
- 830 JVM tests with no device plus 38 instrumented tests on API 26 and 36; released as a signed 2.3 MB APK.

TideFit | TypeScript, Next.js 16, React 19, Supabase, Vercel github.com/UtkarshMitta/tide-fit

The app is Utkarsh Mittal's (UtkarshMitta on GitHub). The audit, the upgrade, the tests, the CI and the launch are mine.

- Security audit of a Next.js trip planner: 15 findings confirmed on the running app (OAuth CSRF, open Supabase policy, secret in a plaintext cookie), 13 fixed.
- Upgraded Next 14 to 16 and React 18 to 19 by hand; npm audit 6 vulnerabilities to 0, tests 0 to 45, CI deploying main to Vercel.
- Launched at tide-fit.vercel.app: private Blob storage, edge rate limit verified at 20 requests then 429, Strava and Google sign-in verified live.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Public leaderboard 0.952. Final ranking pending, competition ends 2026-09-29.
- **Gemma 4 Developer Agent, Kaggle (Google DeepMind)**, 2026. In progress, ends 2026-12-02.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Systems: CMake, gRPC, SQLite, WebAssembly, Android (Jetpack Compose, Room), Next.js, React, FastAPI, Node.js, REST APIs, MCP

Tools: Git, GitHub Actions, Docker, AWS (CDK, App Runner, DynamoDB), Vercel, Linux, WSL2, Tailscale

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Streamlit, Altair, matplotlib

ACTIVITIES

- Treasurer, OWASP Student Chapter, IIIT Delhi